Bachelor of Information System

IS2204 - Information System Security

Lecture – 7 : Credit Card Payment

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Cybersecurity Protocols

- **PGP :** E-Mail, RFC2015
- **S/MIME :** E-Mail, RFC2311,RFC2634

Email Security

- ~~**S-HTTP :** Web Browsing, RFC2660~~
- ~~**PCT :** TCP/IP-level Encryption~~
- ~~**SSL :** TCP/IP-level Encryption, Netscape~~
- **TLS :** TCP/IP-level Encryption, RFC2246

Web Security

- ~~**Cybercash :** Electronic Transactions, RFC1898~~
- ~~**SET :** Secure Electronic Funds Transactions~~
- **BitCoin:** Blockchain CryptoCurrencies

Payment Security

- **SSH :** Remote Login
- **DNSSEC :** Domain Name System, RFC2065
- **IPSec :** Packet-Level Encryption, RFC2401
- **Tor:** Onion Routing Protocol

Network Security

Characteristics of Payments

Properties :

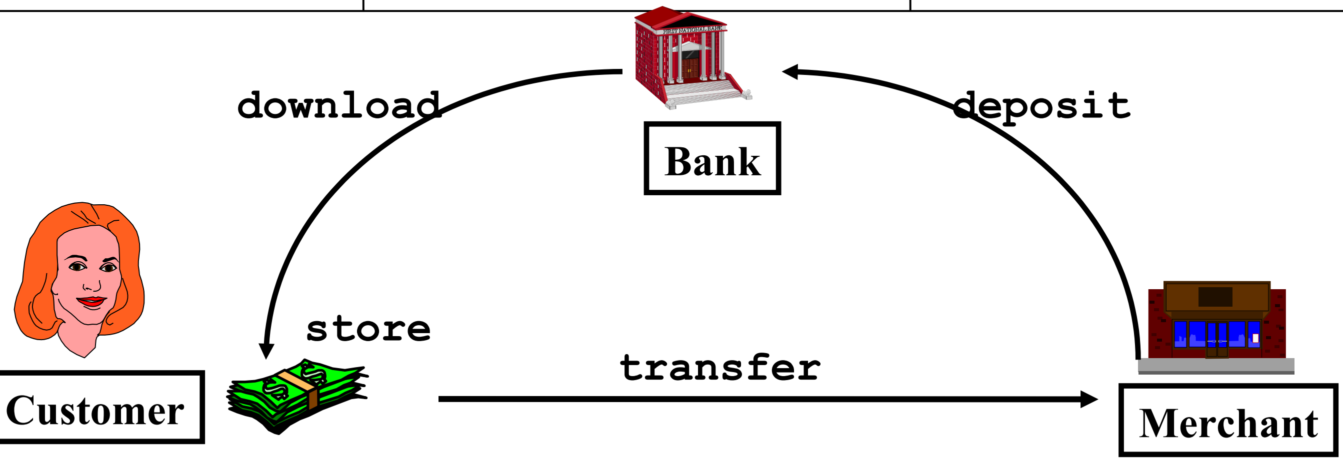
- **where is the money (authorization)**
- **time of payment vs. time of order/shopping**

Characteristics of payment methods :

Type of payment	Money	Time
Cash	with Customer	at Purchase
Debit card	in Bank	at Purchase
Credit card	in Bank	after Purchase
Invoice	in Bank	after Purchase
Pre-paid	with Merchant	before Purchase
Subscription	with Merchant	before Purchase

Internet transactions

Type of payment	Money	Time
Cash	***with Customer***	at Purchase
Debit card	in Bank	at Purchase
Credit card	in Bank	after Purchase
Invoice	in Bank	after Purchase
Pre-paid	with Merchant	before Purchase
Subscription	with Merchant	before Purchase



Internet Transactions

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Debit card	*in Bank*	at Purchase
Credit card	*in Bank*	after Purchase
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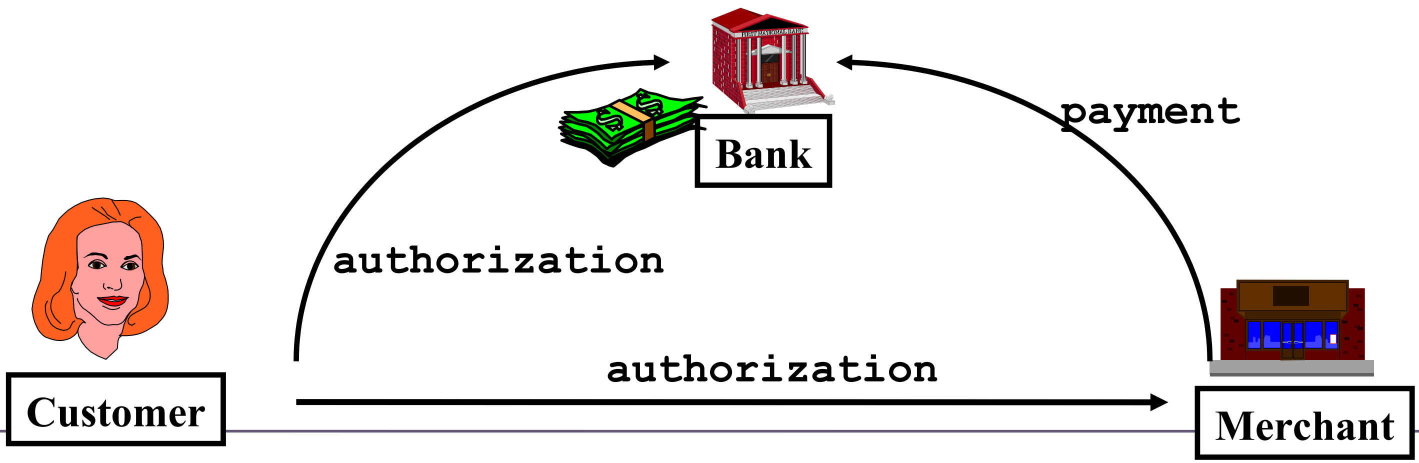

Internet Transactions

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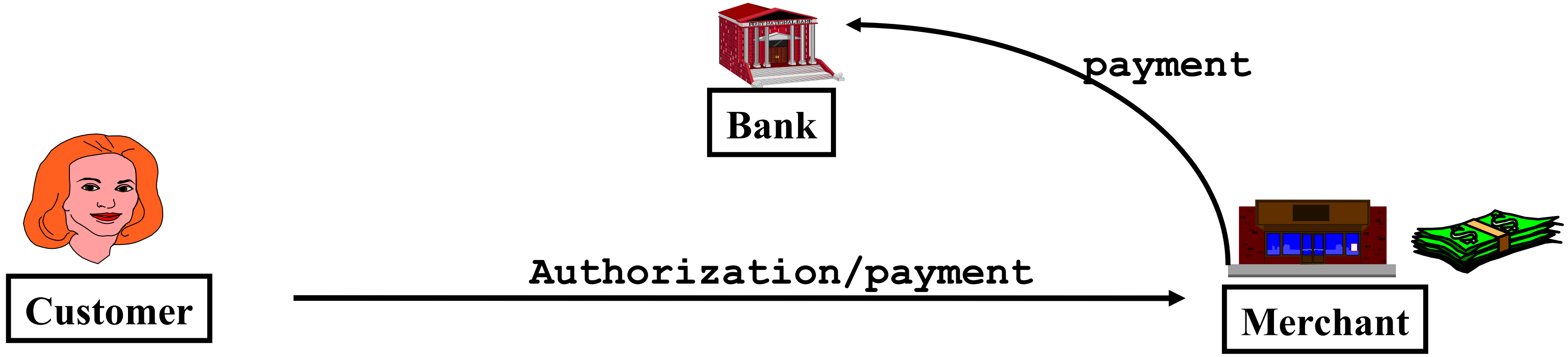


Types of Digital Payments

Characteristics :

- where is the money (authorization)
- time of payment vs. time of order

Types of digital payments :

1. Digital cash
2. Stored money (micropayments)
3. eCheck
4. eMoney order
5. Debit payment
6. Credit payment
7. Invoice / payment order
8. At delivery (pay-per-view)
9. Subscription
10. Cryptocurrencies

Electronic credit and debit

- Standard authentication, confidentiality, and non-repudiation techniques can be used
  - Asymmetric encryption and certificates
- Framework must take into account different institutions involved
- Example: Secure Electronic Transactions (SET) of Visa/Mastercard

Risk in using Credit cards

- Customer uses a stolen card or account number to fraudulently purchase goods or service online
- Family members use bankcard to order goods/ services online, but have not been authorized to do so.
- Customer falsely claims that he or she did not receive a shipment
- Hackers find the ways into an e-commerce merchant's payment processing system and then issue credits to hacker card account numbers.

Risk in using Credit cards

Extra protection when there's no card

Card-not-present (CNP) merchants must take extra precaution against fraud exposure and associated losses. Anonymous scam artists bet on the fact that many Visa fraud prevention features do not apply in this environment. Follow these recommendations to help prevent fraud in your card-not-present transactions.

Quick steps to ensure against CNP fraud

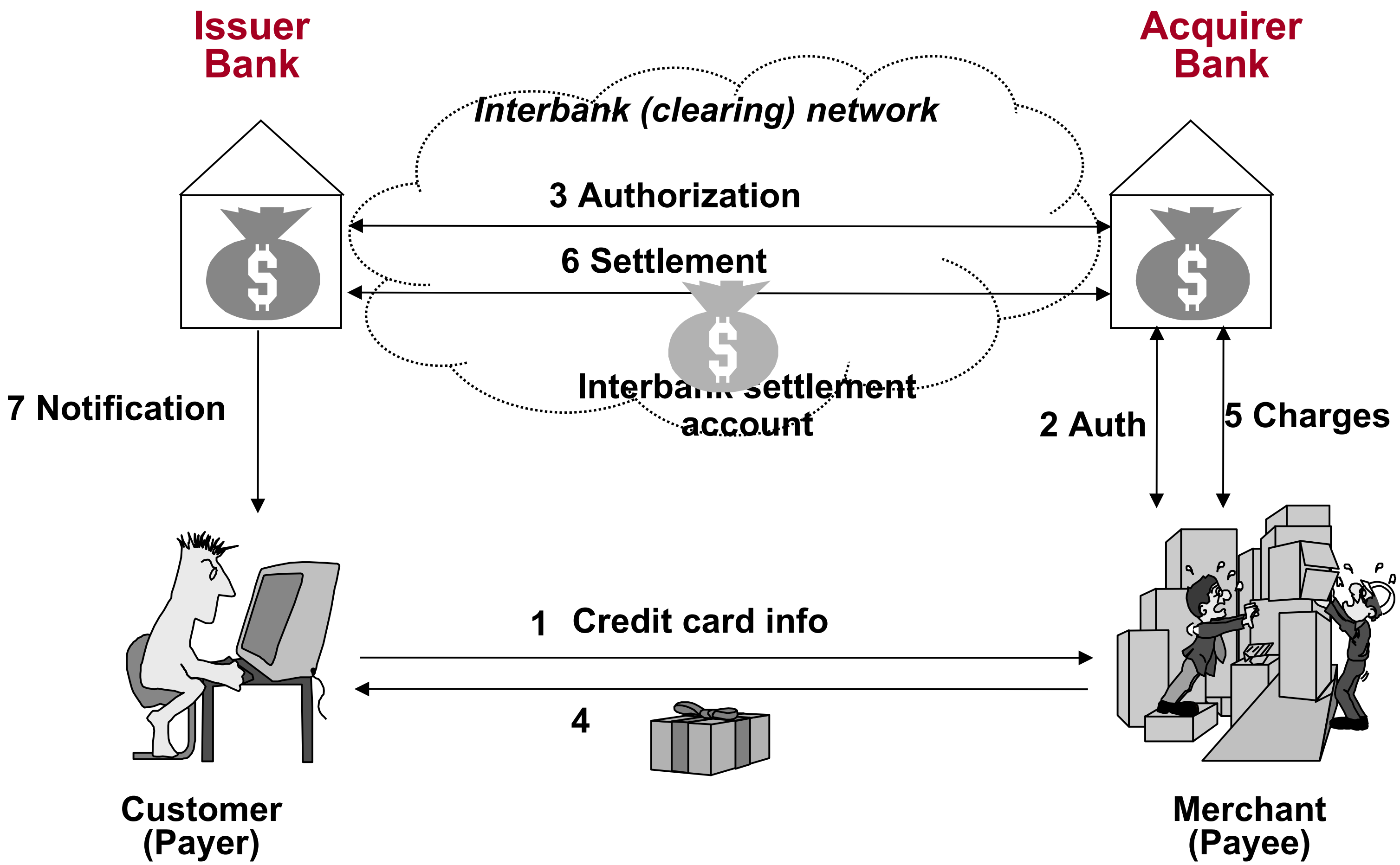
Obtain an authorization.

Verify the card's legitimacy:

Ask the customer for the card expiration date, and include it in your authorization request. An invalid or missing expiration date might indicate that the customer does not have the actual card in hand.

Use fraud prevention tools such as Visa's Address Verification Service (AVS), Card Verification Value 2 (CVV2), and Verified by Visa.

Credit/Debit Card Payments



Credit Card Protocols

- SSL 1 or 2 parties have private keys
- TLS (Transport Layer Security)
  - IETF version of SSL

VERY IMPORTANT. USAGE INCREASING

- *i*KP (IBM) i parties have private keys
- SEPP (Secure Encryption Payment Protocol)
  - MasterCard, IBM, Netscape based on 3KP
- STT (Secure Transaction Technology)
  - VISA, Microsoft
- SET (Secure Electronic Transactions)
  - MasterCard, VISA all parties have certificates

OBSOLETE

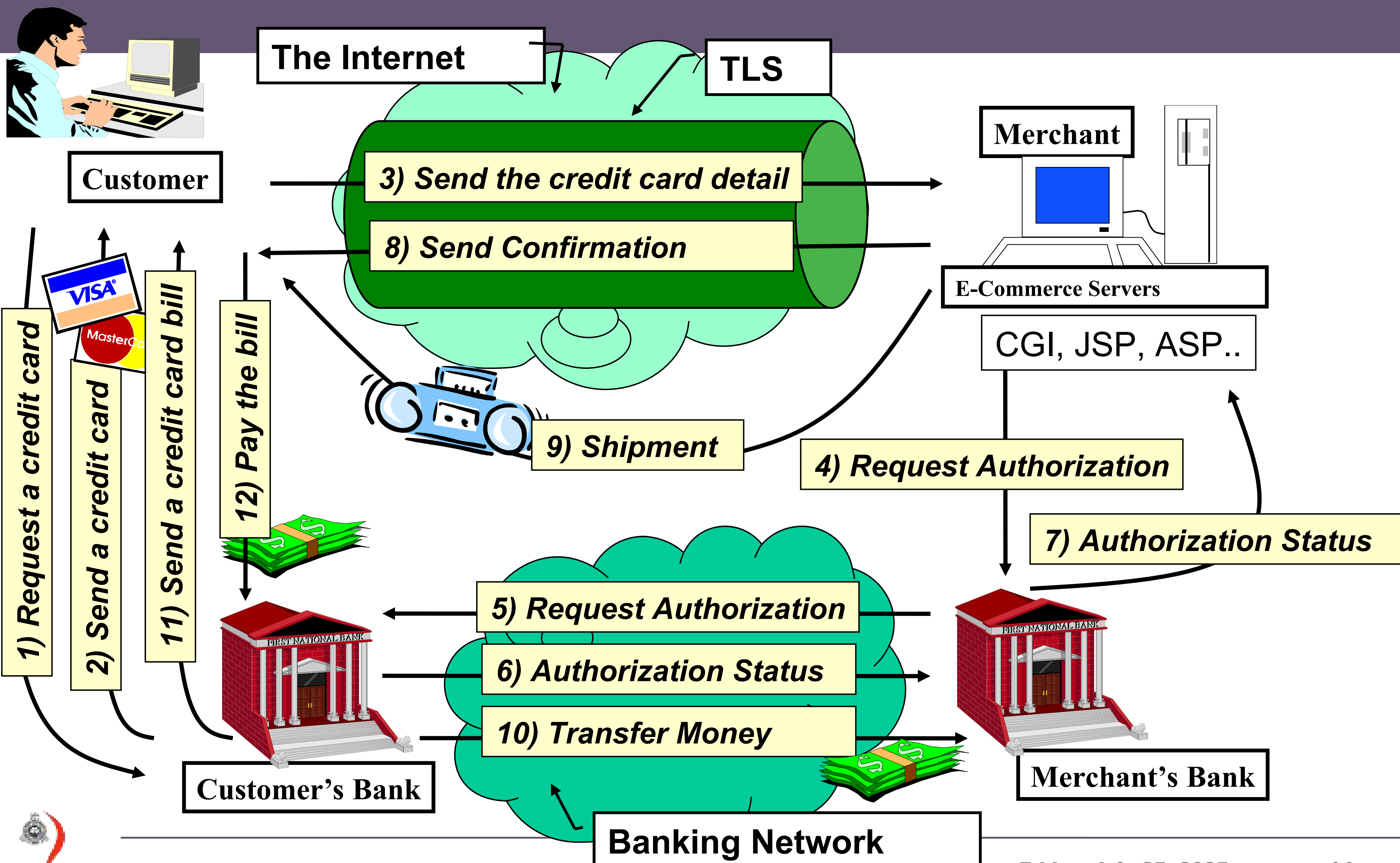
- 3D Secure

VERY SLOW ACCEPTANCE

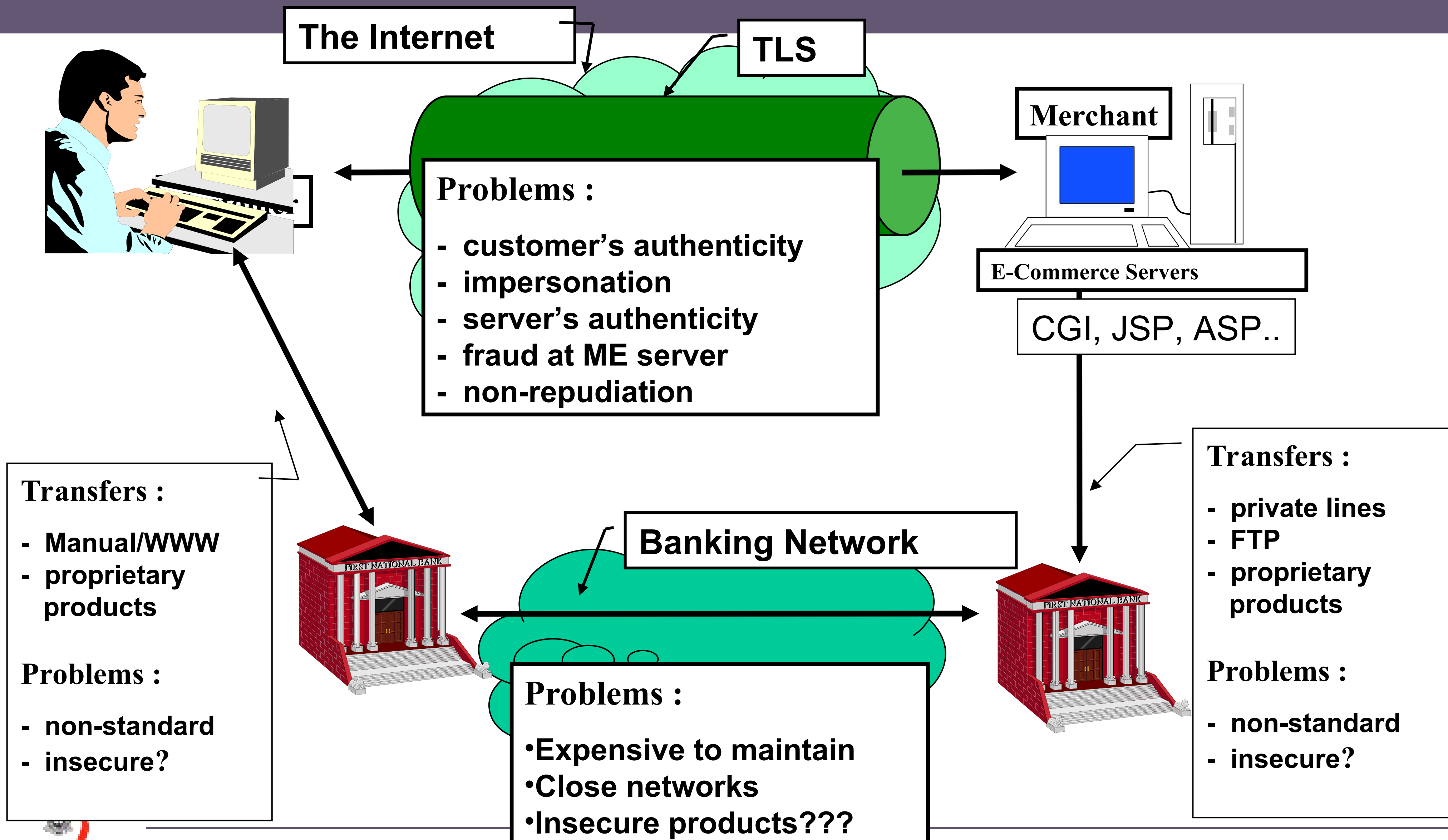
TLS (Transport Layer Security)

- NOT a payment protocol -- can be used for any secure communications, like credit card numbers
- TLS is a secure data exchange protocol providing
  - Privacy between two Internet applications
  - Authentication of server (authentication of browser optional)
- Uses enveloping: RSA used to exchange AES keys
- TLS Handshake Protocol
  - Negotiates symmetric encryption protocol, authenticates
- TLS Record Protocol
  - Packs/unpacks records, performs encryption/decryption
- **Does not provide non-repudiation**

Internet Transactions (Insecure?)



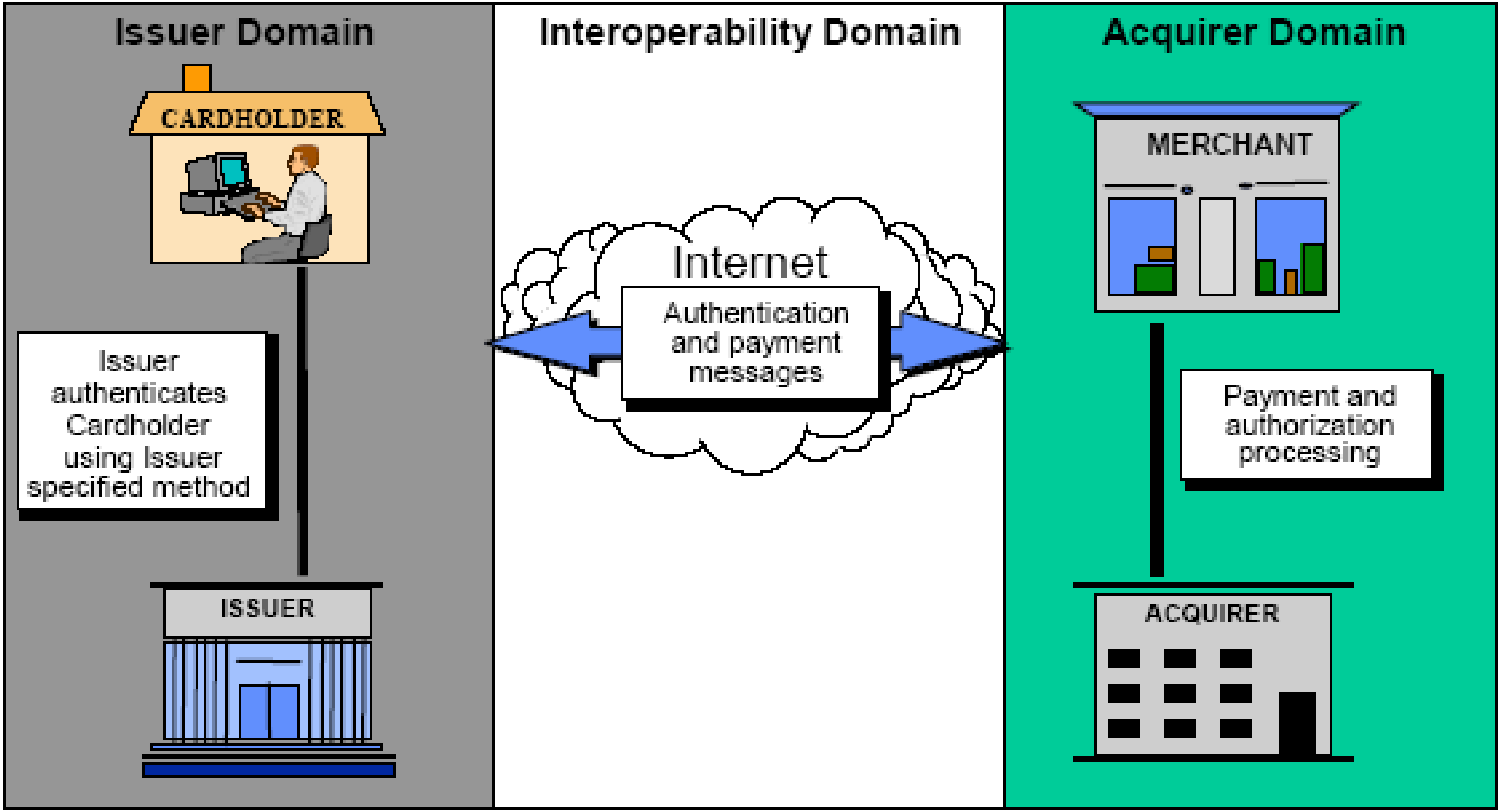
Secure Credit Card Payments (TLS)



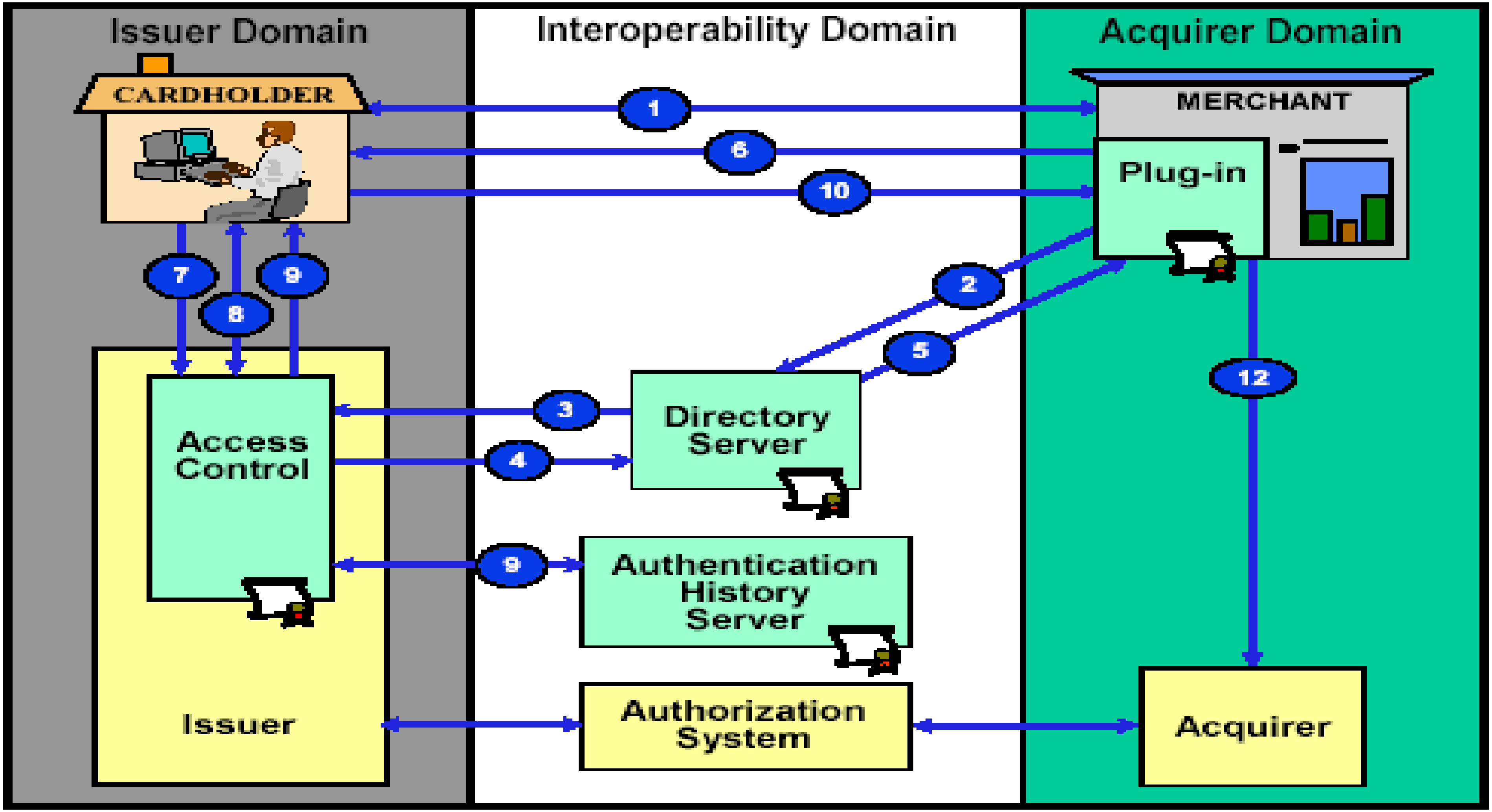
3-D Secure

- Idea: authenticate user without a certificate
- Requires the user to answer a challenge in real-time
- Challenge comes from the issuing bank, not the merchant
- Issuing bank confirms user identity to merchant

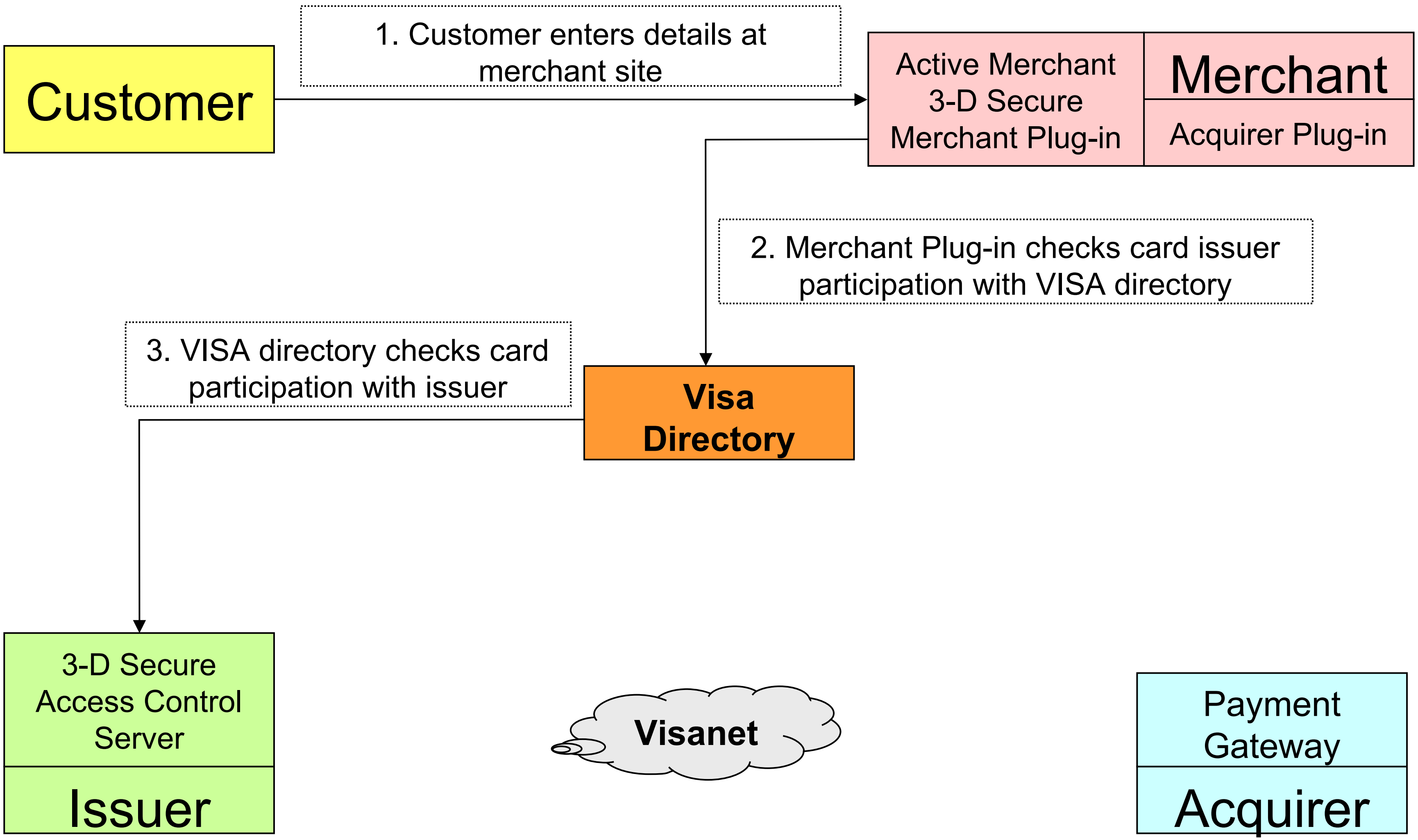
Overview of 3-D Secure



How Does 3-D Secure Work

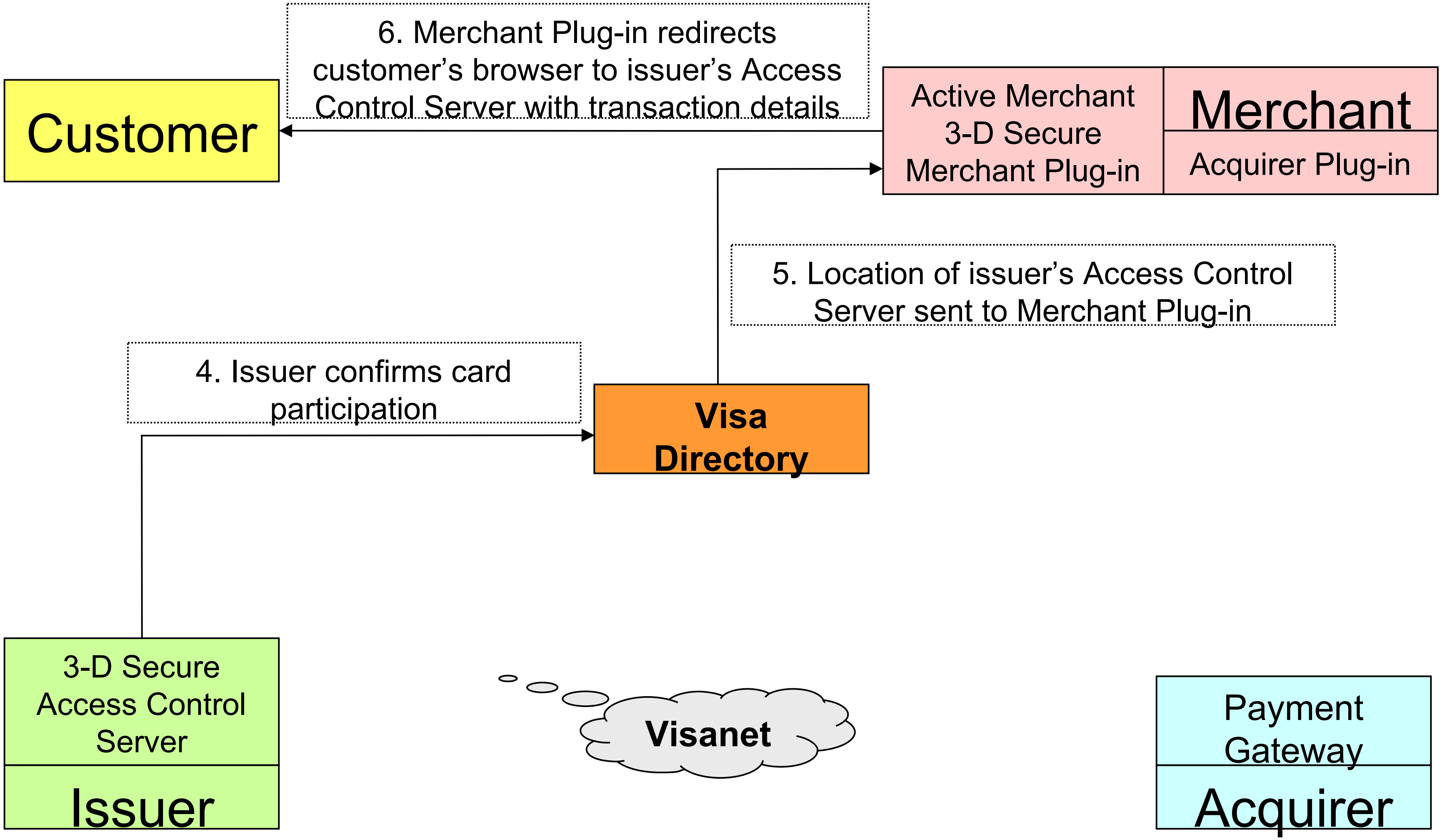


3-D Secure (1)



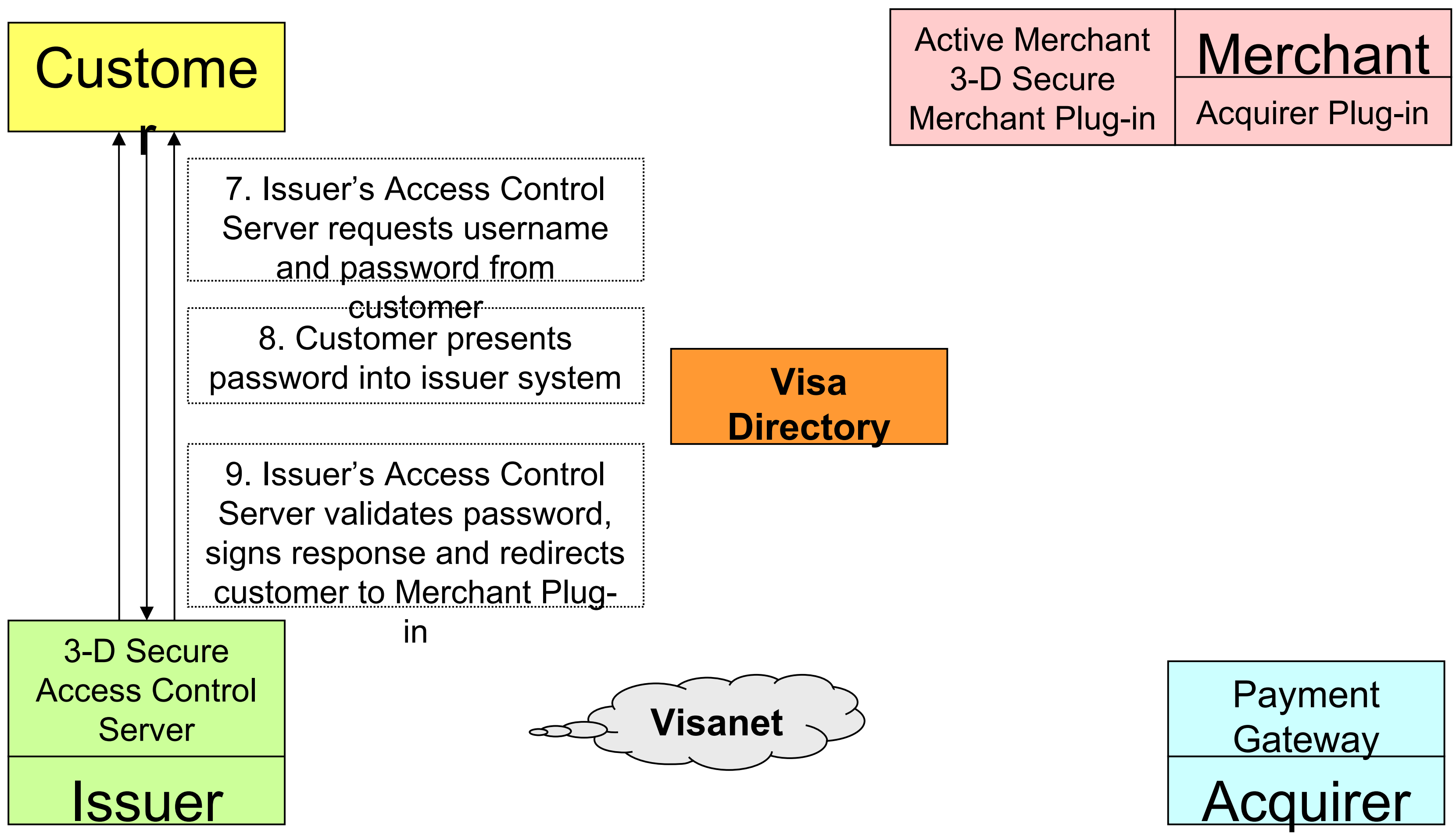
SOURCE: KMIS

3-D Secure (2)



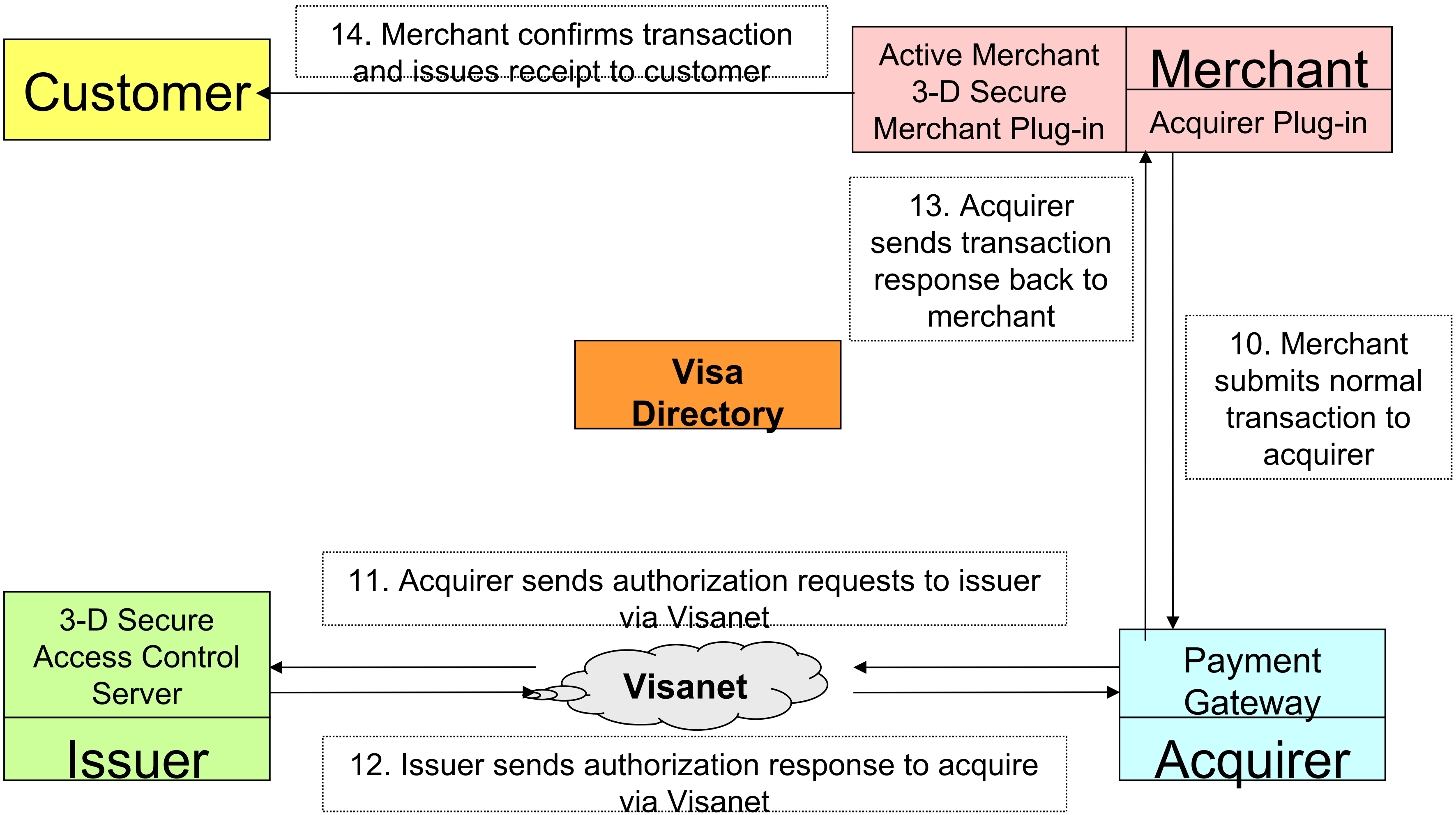
SOURCE: KMIS

3-D Secure (3)



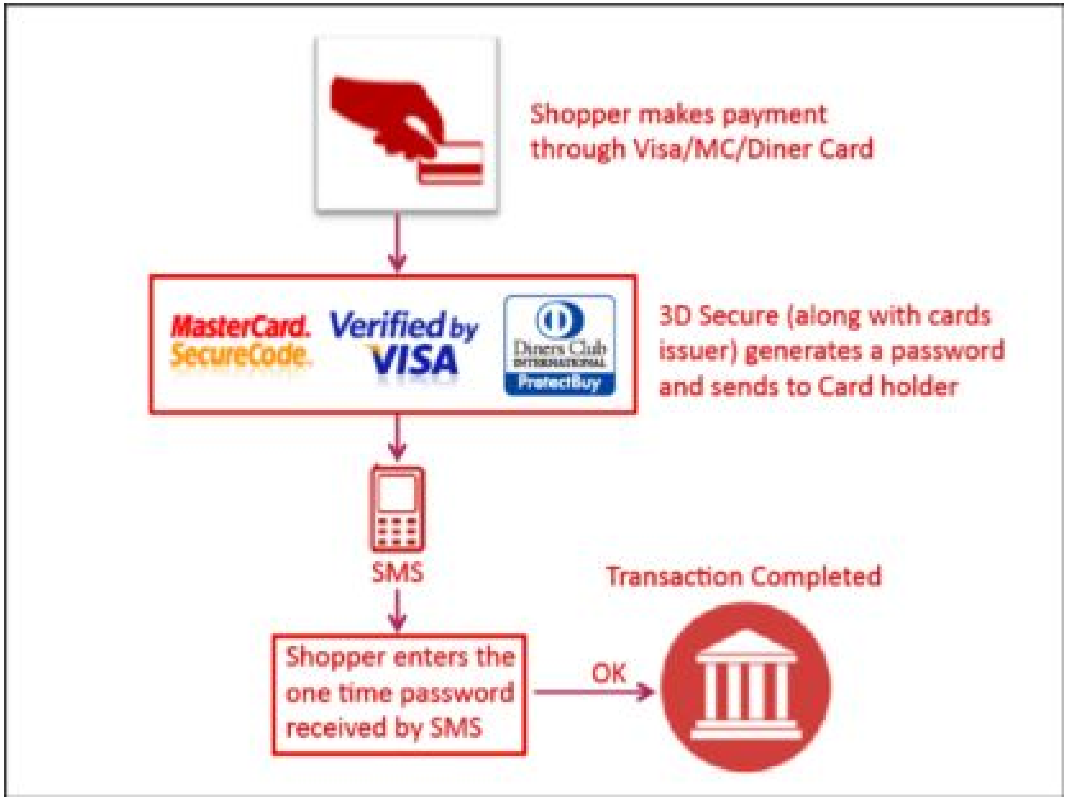
SOURCE: KMIS

3-D Secure (4)


14. Merchant confirms transaction and issues receipt to customer
Customer
Active Merchant 3-D Secure Merchant Plug-in
Merchant
Acquirer Plug-in
13. Acquirer sends transaction response back to merchant
10. Merchant submits normal transaction to acquirer
Visa Directory
11. Acquirer sends authorization requests to issuer via Visanet
3-D Secure Access Control Server
Issuer
Visanet
Payment Gateway
Acquirer
12. Issuer sends authorization response to acquire via Visanet

3-D Secure with OTP

- **Step 1** – Make an online purchase.
- **Step 2** – Enter your card details in the Payment section.
- **Step 3** – An SMS with a one-time password (OTP) will be sent to your registered mobile device.
- **Step 4** – Enter the one-time password (OTP) and complete your transaction.



Features of 3-D secure

- Payment Authentication
  - Issuers to verify that the person involved in e-commerce is a authorized cardholder.
  - Improved transaction performance to benefit all participants
  - Increase consumer confidence
- Support variety of Internet access devices
  - Personal Computer
  - Mobile Phones
  - Personal Digital Assistants

Benefit of 3-D Secure

- Benefit for Cardholder
  - Increased Customer confidence
  - No Application software is needed
  - Easy to use
- Benefit for Merchants
  - Ease of integration into merchant system
  - Reduce risk of fraudulent transaction
  - Decrease in disputed transactions

eCheck

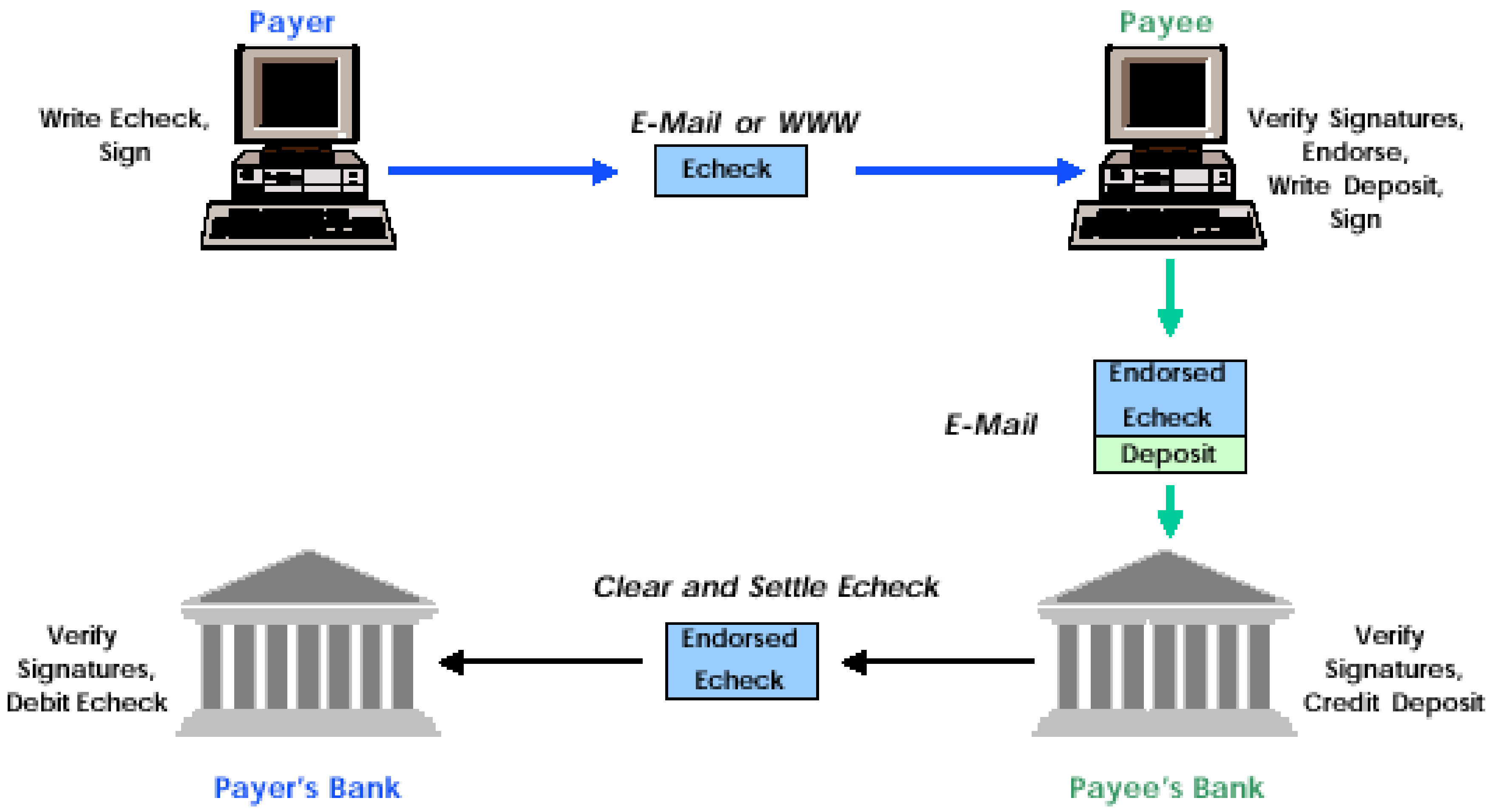
What are electronic checks?

•Checks, without the paper
•***Bank*** payments, secure enough for the Internet
•Digitally signed promises to pay

Significance

•New payment alternative for business commerce
•It's real, and working today
•It's interoperable, with multiple providers
•It fits and enhances existing business practices
•It extends checking into the 21st century

eCheck

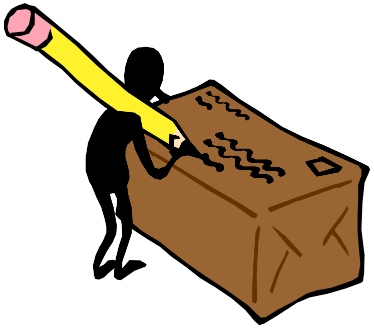

Payer
Write Echeck, Sign
E-Mail or WWW
Echeck
Payee
Verify Signatures, Endorse, Write Deposit, Sign
E-Mail
Endorsed Echeck
Deposit
Clear and Settle Echeck
Endorsed Echeck
Verify Signatures, Debit Echeck
Verify Signatures, Credit Deposit
Payer's Bank
Payee's Bank

eMoney Order

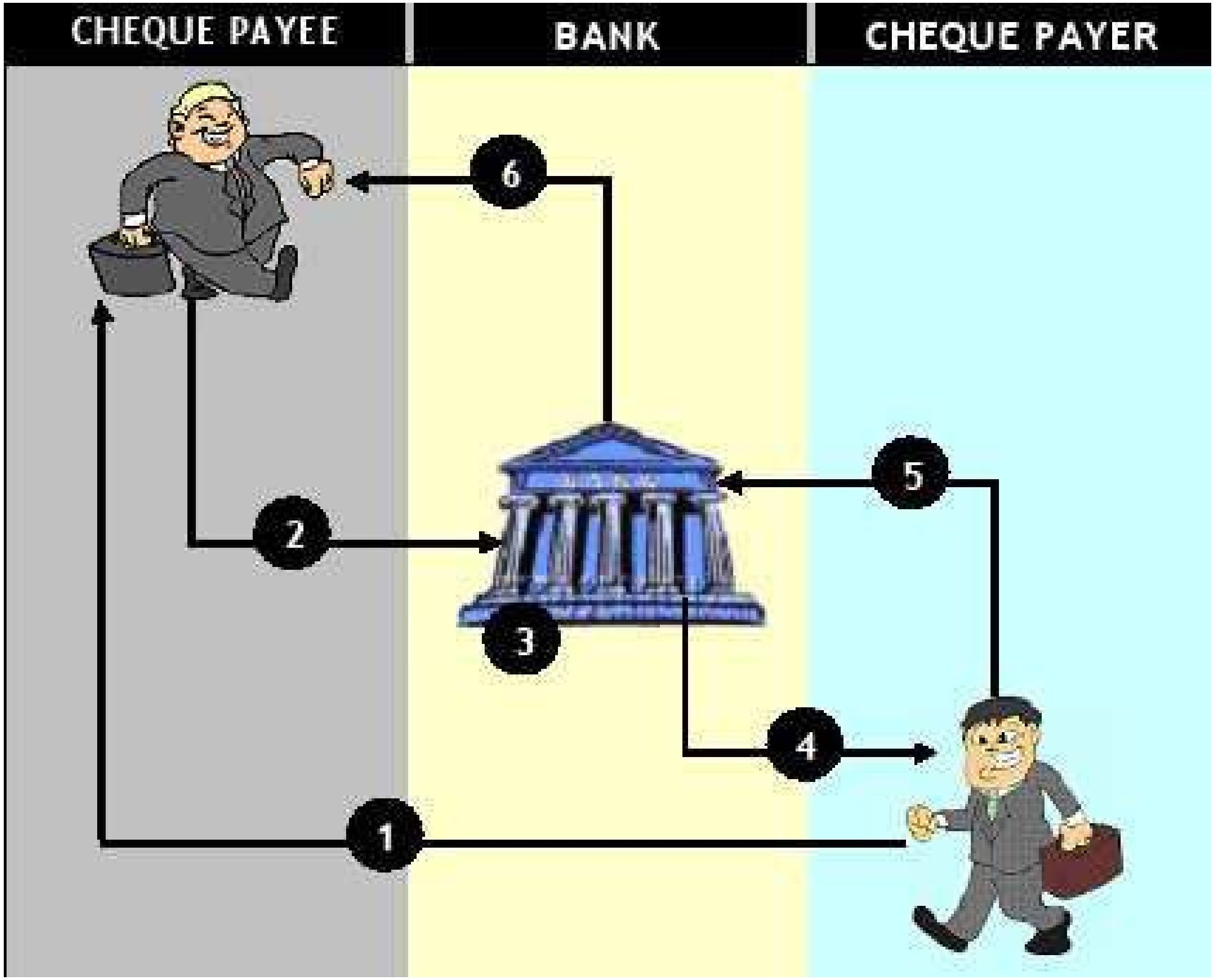
- Advantages
  - Easy to access
    - A large number of post offices & banks are available all over the country.
  - Easy to understand
    - Not a completely new system. This is an enhancement to the existing system.
  - Save the money within the country
    - In using credit cards, commission has to paid to International Credit Card companies

eMoney Order : How does it work?

- A persons goes to the nearest post office
- Pay the required amount and buy the eMoney order
- Send the number in the eMoney order together with other details in the web form.



Trusted Cheque protocol (TCP)

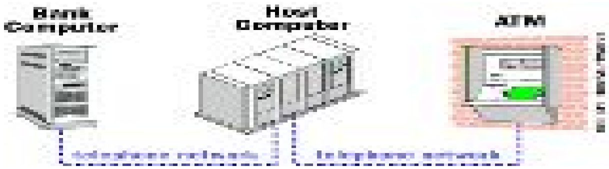


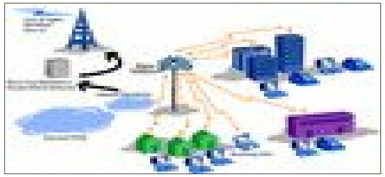
M-ATM (Mobile ATM)

Provide efficient secured solution for the barriers of using ATM with support of mobile phones

ATM Machine → **Mobile Phone**

ATM Card → **SIM Card Java Card)**





Bank Network → **Mobile Network**

Problems

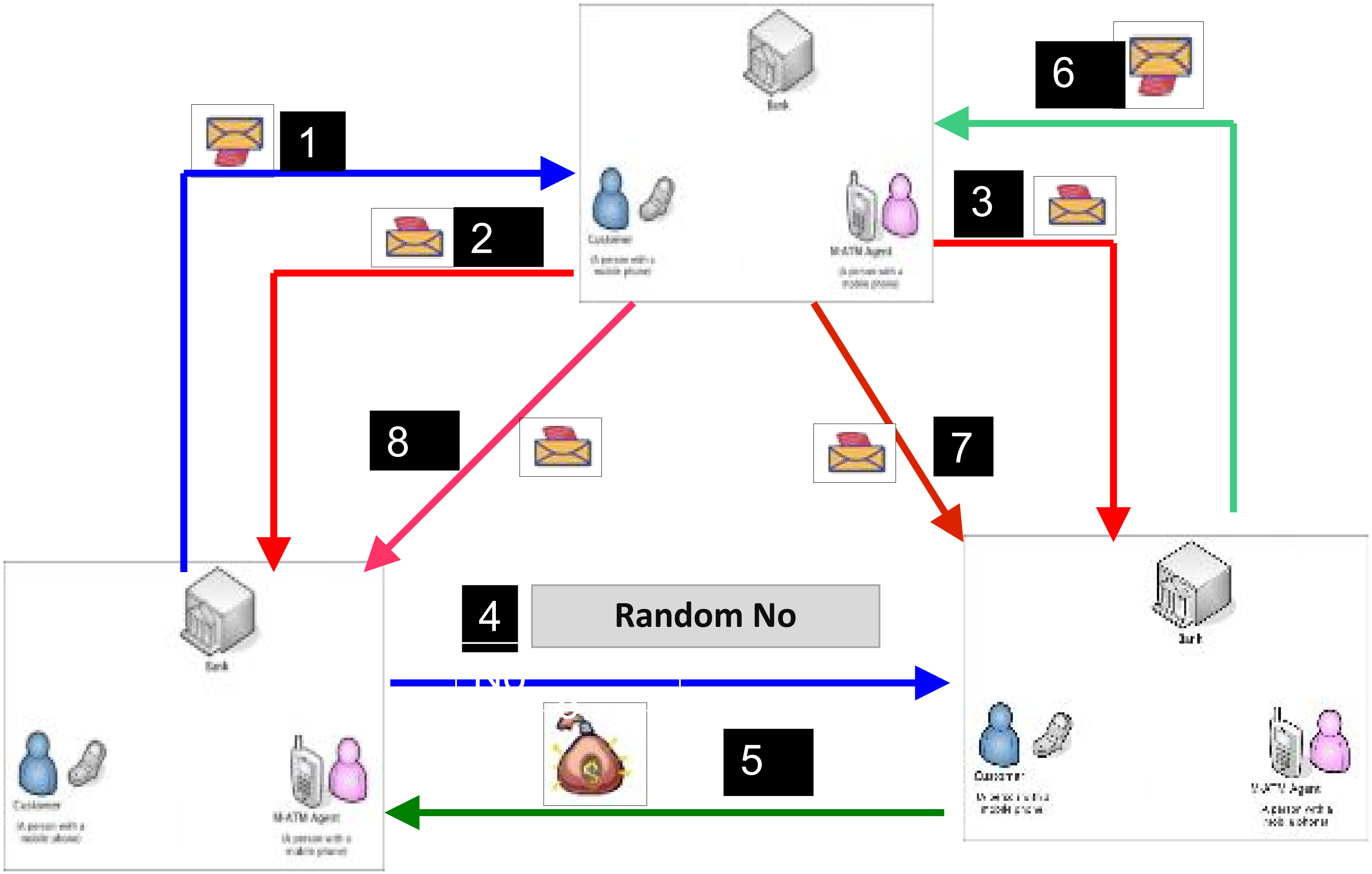
Bank's Perspective

* Secure connection is needed
* Skilled staff to maintain ATM
* Security risk
  - protect against theft
* Initial capital of deploying ATM
  - very high

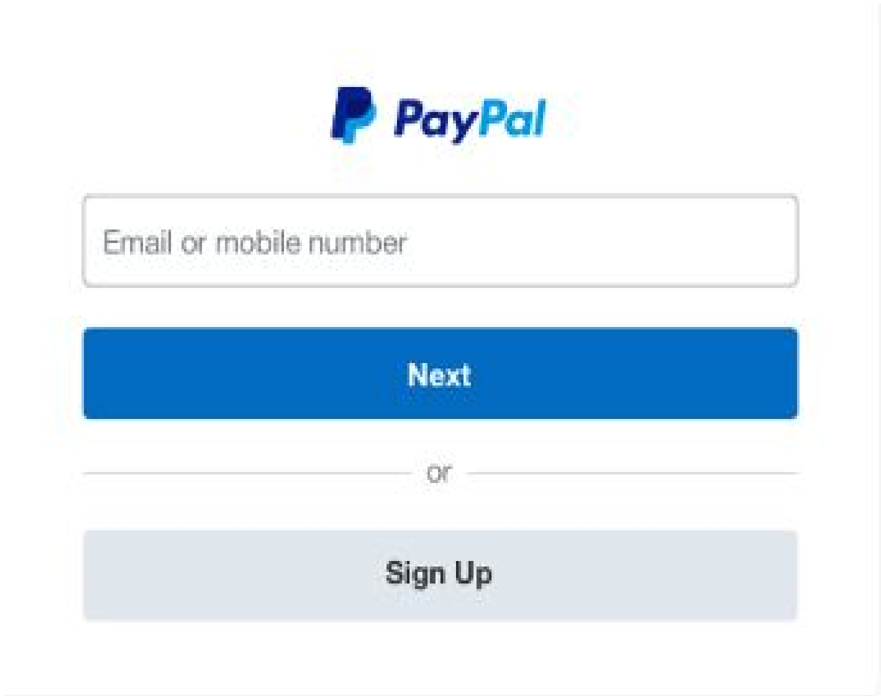
Customer's Perspective

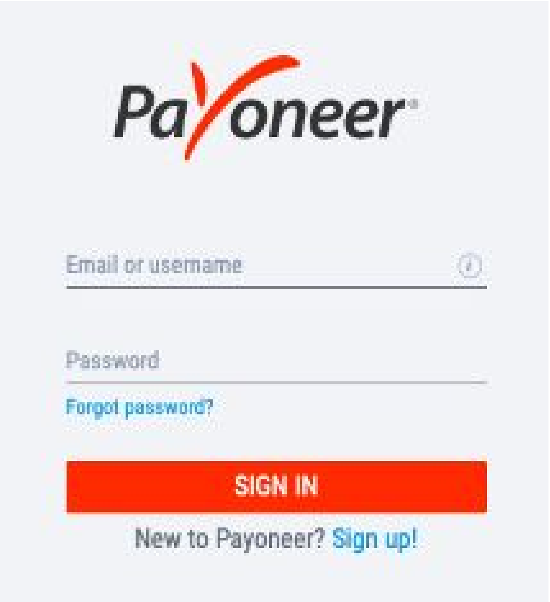
* Not very common in rural areas.
* Users have travel more to ATM
* Security risk
* Special plastic card involved in transaction

Overall Architecture



PayPal vs Payoneer





Discussion